

1. Project Overview

- Project Name: Nutritional AI Agent
- Core Scope: Autonomous and personalized daily dietary planning and recipes.
- UN SDG Alignment: Goal 3 (Good Health and Well-being).
- Technologies: Streamlit, LangGraph (StateGraph), LangChain, Groq API (Llama 3.3 70B).

2. Introduction & Relevance to SDG 3

Good Health and Well-being (UN SDG 3) emphasizes healthy lifestyles and disease mitigation. Metabolic health is directly determined by dietary intake, yet obesity and malnutrition affect millions worldwide. Standard tools fail because manual logs require high cognitive friction and do not provide personalized, context-aware suggestions. This project implements a conversational, agentic solution that autonomously calculates nutrient requirements and designs complete daily recipe plans.

3. Problem Statement & System Objectives

- The Objective: Develop an autonomous AI system that extracts personal metabolic metrics (Age, Weight, Goal, Allergies) and compiles highly optimal meal structures.
- Specific Technical Goals:
 - Automate user feature processing through a secure Streamlit UI sidebar.
 - Coordinate nodes in a sequential StateGraph to guarantee clean recipe outputs.
 - Format recipes into structured target calorie breakdown tables (Breakfast, Lunch, Dinner).

4. Technical Methodology & Solution Architecture

The system utilizes a modern agentic AI architecture with a sequential dual-agent LangGraph workflow:

1. User Interface (Streamlit): Collects physical parameters and fitness targets safely in the sidebar.
2. State Management (TypedDict): Passes feature attributes dynamically across nodes.
3. Extraction Node (Dietary Researcher): Prompts Groq API (Llama-3.3-70b-versatile) to run clinical diet research and output ingredients, portion sizes, and health rationales.
4. Formatting Node (Content Writer): Re-writes raw text into clean, markdown-compliant tables.
5. Output Render: Main content area updates asynchronously showing the complete recipes.

5. Development Tools & Future Scope

- Core Stack: Python, Streamlit, LangGraph, LangChain, Groq API (Llama 3.3 70B).
- IBM Cloud & Watson Integration Concept: Uses IBM Cloud Code Engine for serverless Streamlit hosting and IBM Cloud Object Storage for logging recipes.
- Future Work: Integrate wearable sensors (e.g. Fitbit/Apple Watch) for dynamic heart-rate/activity-adjusted calorie models, use Vision LLMs to log meal photographs, and train Watson Studio predictors for custom BMR estimation.